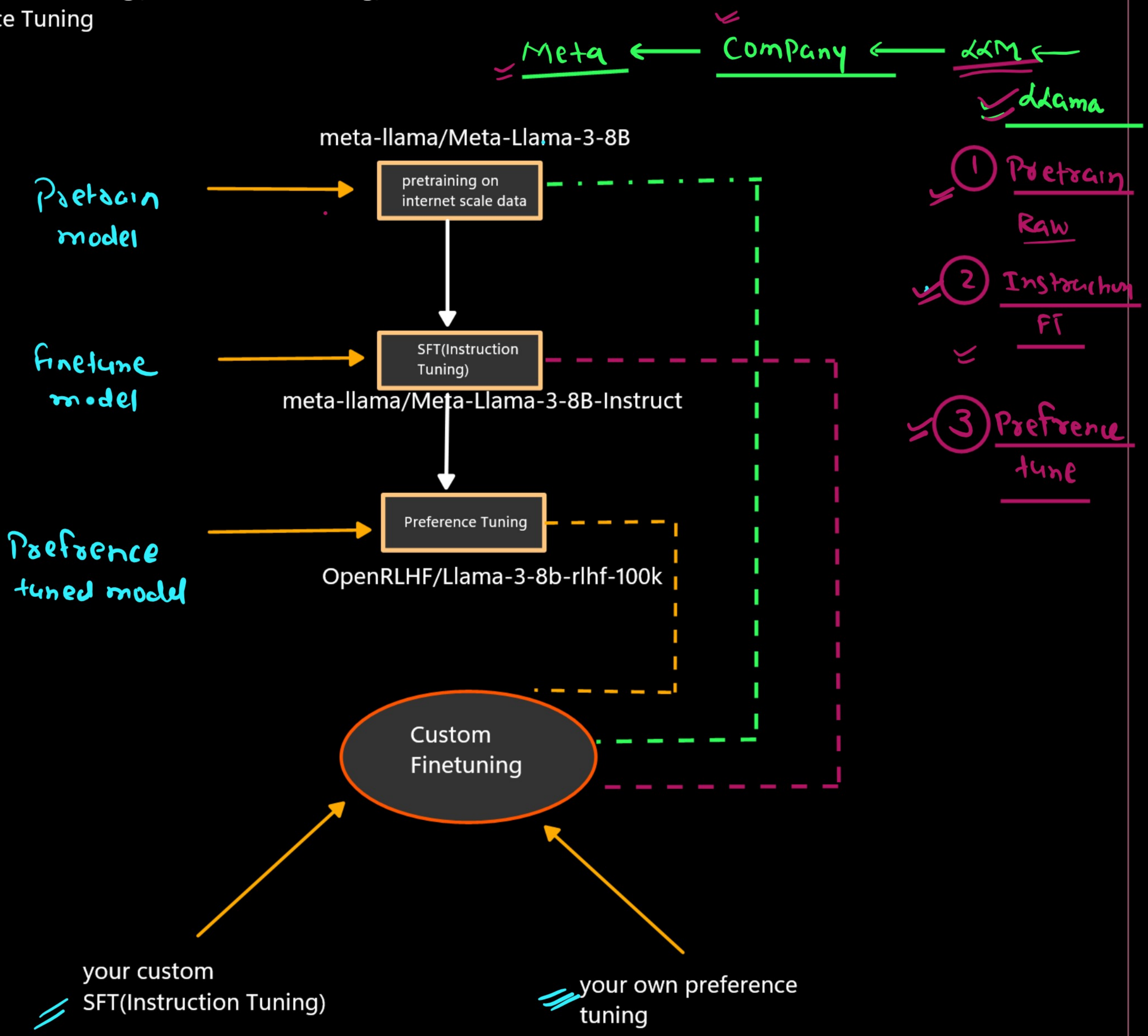
Fine-tuning

Fine-tuning is the process of taking a already trained model and training it further on domain-specific or task-specific data so that it performs better for a particular use case.

LLM Training Pipeline
1. Pretraining of model
2. Supervised Finetuning(Instruction Finetuning)
3. Preference Tuning



1. hugging face Modules
2. unsloth

1. FULL Finetuning
2. Partial Finetuning(PEFT)

LLM → 1 Meta
2 Mistral

{ train the model on your own data }

customtuning

1 Pretraining ← RAW

2 SFT | Instruction ← I/P ⟷ O/P

3 Preference tuning

meta-llama/Meta-Llama-3-8B

↓

meta-llama/Meta-Llama-3-8B-Instruct
(how to follow the instruction)

usecase: Pharama --> document on molecule study
1. to train a llm from scratch it is difficult
2. i will take any opensource model i will download it on my owm server
3. and then i will finetune that

my requirement was just to generate the data not the conversation AI/Chatbot/QA

my requirement was just to generate the data
meta-llama/Meta-Llama-3-8B --> i have trained(finetuning) on my own pharama datset

conversation AI/Chatbot/QA
meta-llama/Meta-Llama-3-8B-Instruct --> further i will finetuning it on my own dataset
IP--OP

model → Parameter (weight | bias) (trained)

transformer → Self Attention ← Q | K | V ← weight

FFNN ← weight

1. Pretraining (Unsupervised | Self supervised learning)

2. SFT / IFT → Supervised finetuning (I/P ⟷ O/P)

Parameter level

① Huge GPU Power ② huge infra

1. ~~Full finetuning~~ ← Retrain all the Parameter (weight & bias)

2. Partial finetuning ← Subset of Parameter

Two method

1. OLD School method

CV
- CNN → VGG | Resnet ...
- BERT
- BART
- T5

retrain

1. Freeze all the layer train only last O/P layer

2. Freeze some starting layer & retrain the remaining last layer

(Latest ← LLM ← huge architecture)

Latest

2. PEFT ← Small Infra

Single GPU | small VRAM

Parameter efficient FT

Low Rank Adaption — 1. LORA → QLORA

Quantized (Float → small float / int)

ADAPTER

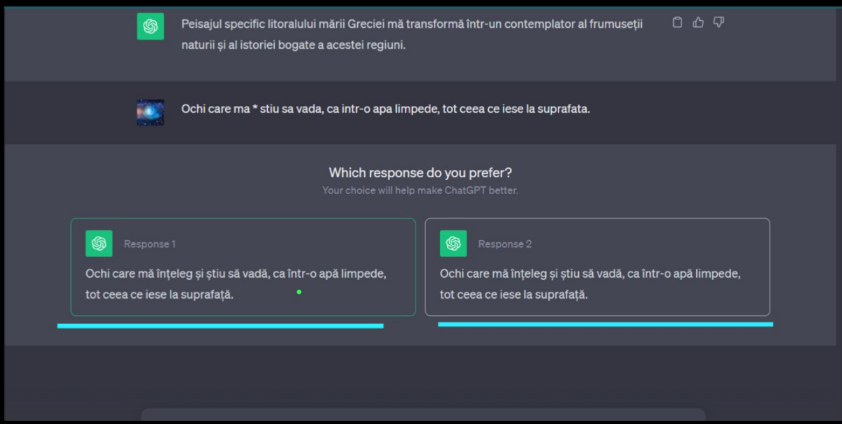
(weight-decomposition) — 2. DORA

3. BitFit

4. IA^3

X

3 Prefrence alignment



collect

Again we train

Retrain our model

1 $\boxed{\text{RLHF}}$ → OpenAI → RL → DRL → $\boxed{\text{PPO}}$

framework

(2 GRPO)

3 $\boxed{\text{DPO}}$ (~~RL~~) → (4 ORPO)